What this instability is about?

We find divergence of $\chi(\vec{Q},0) = \dfrac{\chi_0(\vec{Q},0)}{1+U\chi_0(\vec{Q},0)}$

with $\vec{Q} = \left(\frac{\pi}{a}, \frac{\pi}{a}\right)$

$\Rightarrow$ expect antiferromagnetic order $\langle \vec{S}(\vec{r}) \rangle = \vec{m} \cos(\vec{Q}\cdot\vec{r})$.

$$H_{int} = U \sum_r \left(n_{r\uparrow} - \tfrac{1}{2}\right)\left(n_{r\downarrow} - \tfrac{1}{2}\right)$$

At $\frac{1}{2}$-filling $\langle n_{r\sigma} \rangle = \frac{1}{2}$.

Mean-field : $\left.\begin{array}{l} \langle n_{r\uparrow} - \frac{1}{2} \rangle = (-1)^r m_0 \\ \langle n_{r\downarrow} - \frac{1}{2} \rangle = -(-1)^r m_0 \end{array}\right\}$ so that $\langle S_r^z \rangle = \frac{1}{2}\langle n_{r\uparrow} - n_{r\downarrow} \rangle = (-1)^r m_0$

Now we can approximate interactions as

$$H_{int} \approx U \sum_r \left(n_{r\uparrow} - \tfrac{1}{2}\right)\langle n_{r\downarrow} - \tfrac{1}{2}\rangle + \langle n_{r\uparrow} - \tfrac{1}{2}\rangle\left(n_{r\downarrow} - \tfrac{1}{2}\right) =$$

$$= U \sum_r (-1)^r m_0 \left\{ -\left(n_{r\uparrow} - \tfrac{1}{2}\right) + n_{r\downarrow} - \tfrac{1}{2} \right\} =$$

$$= U \sum_r -(-1)^r m_0 \left\{ n_{r\uparrow} - n_{r\downarrow} \right\} \quad (\text{eq. } 1)$$

With this approximation $H = H_0 + H_{int}$

decouples into $H = H_\uparrow + H_\downarrow$

$$H_\uparrow = \sum_k \varepsilon(k)\, c^+_{k\uparrow} c_{k\uparrow} - U m_0 \sum_r (-1)^r c^+_{r\uparrow} c_{r\uparrow}$$

$$H_\downarrow = \sum_k \varepsilon(k)\, c^+_{k\downarrow} c_{k\downarrow} + U m_0 \sum_r (-1)^r c^+_{r\downarrow} c_{r\downarrow}$$

$\uparrow(\downarrow)$ electrons move in a self-consistent potential created by the opposite spin density.

More on mean-field approximation (detail of eq.1 of previous page)

$$H_{int} = U \sum_r m_{r\uparrow} m_{r\downarrow}$$

(denote $m_{r\sigma} = n_{r\sigma} - \frac{1}{2}$ for simplicity)

$$H_{int} \approx U \sum_r \left(\overline{m}_{r\uparrow} + m_{r\uparrow} - \overline{m}_{r\uparrow}\right)\left(\overline{m}_{r\downarrow} + m_{r\downarrow} - \overline{m}_{r\downarrow}\right)$$

$$\cong U \sum_r \Big(\overline{m}_{r\uparrow}\, \overline{m}_{r\downarrow} + \overline{m}_{r\uparrow}\left(m_{r\downarrow} - \overline{m}_{r\downarrow}\right) +$$

$$+ \left(m_{r\uparrow} - \overline{m}_{r\uparrow}\right) \overline{m}_{r\downarrow} + \underbrace{\cancel{\left(m_{r\uparrow} - \overline{m}_{r\uparrow}\right)\left(m_{r\downarrow} - \overline{m}_{r\downarrow}\right)}}_{\text{neglect}} \Big)$$

$$= U \sum_r \left(m_{r\uparrow}\, \overline{m}_{r\downarrow} + m_{r\downarrow}\, \overline{m}_{r\uparrow} - \overline{m}_{r\uparrow}\, \overline{m}_{r\downarrow} \right)$$

Observe that $\overline{m}_{r\sigma} = \langle m_{r\sigma} \rangle = \begin{cases} (-1)^r m_0, & \sigma = \uparrow \\ -(-1)^r m_0, & \sigma = \downarrow \end{cases}$

$$\Rightarrow H_{int} = U \sum_r -(-1)^r m_0 \left(m_{r\uparrow} - m_{r\downarrow}\right)$$

$$- (-1)^r m_0 \cdot \left(-(-1)^r m_0\right) =$$

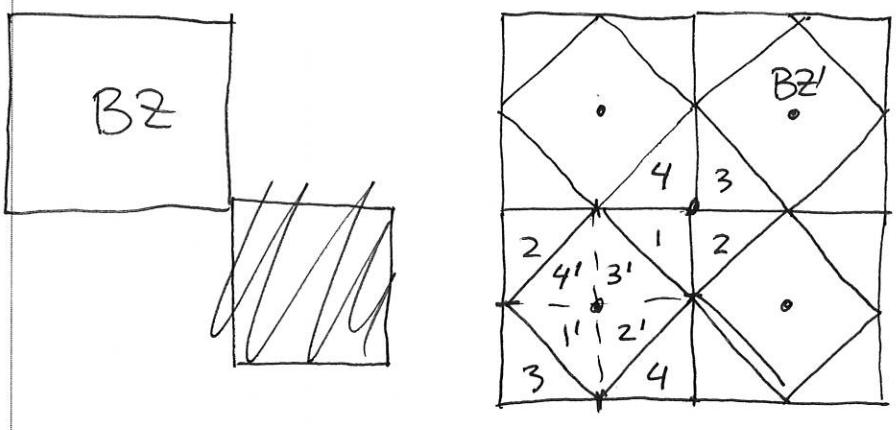
$$= \underbrace{U \sum_r m_0^2}_{N U m_0^2} + U \sum_r -(-1)^r m_0 \left(n_{r\uparrow} - n_{r\downarrow}\right)$$

N = number of sites

But $\sum_r (-1)^r c_r^\dagger c_r = \sum_r e^{i\vec{Q}\cdot\vec{r}} \frac{1}{V} \sum_{k_1 k_2} e^{-i\bar{k}_1\cdot\bar{r}} e^{i\bar{k}_2\cdot\bar{r}}$

$c_{k_1}^\dagger c_{k_2} = \sum_k c_{k+Q}^\dagger c_k$

$\vec{k}_1 = \vec{k}_2 + \vec{Q}$, $\vec{Q} = \frac{\pi}{a}(1,1)$



$$\sum_{k\in BZ} \varepsilon(k) c_k^\dagger c_k = \sum_{k\in BZ'}{}' \varepsilon(k) c_k^\dagger c_k + \sum_{k\in 1}() + \sum_{k\in 2}() +$$

$$+ \sum_{k\in 3}() + \sum_{k\in 4}()$$

$\vec{4} = \vec{4}' + \left(-\frac{\pi}{a}, -\frac{\pi}{a}\right)$

$\vec{3} = \vec{3}' + \left(\frac{\pi}{a}, \frac{\pi}{a}\right)$

$\vec{2} = \vec{2}' + \left(-\frac{\pi}{a}, \frac{\pi}{a}\right)$

$\vec{1} = \vec{1}' + \left(\frac{\pi}{a}, \frac{\pi}{a}\right)$

for example, $\sum_{k\in 1} \varepsilon(k) c_k^\dagger c_k = \sum_{k'\in 1'} \varepsilon(\bar{k}'+\bar{Q}) c_{k'+Q}^\dagger c_{k'+Q}$

$\vec{k} = \vec{k}' + \vec{Q}$

(but $\varepsilon(\bar{k}'+\bar{Q}) = -\varepsilon(k')$) $= \sum_{k'\in 1'} -\varepsilon(k') c_{k'+Q}^\dagger c_{k'+Q}$

$$\Rightarrow \sum_{k\in BZ} \varepsilon(k)\, c_k^+ c_k = \sum_{k^*\in BZ'} \varepsilon(k)\, c_k^+ c_k$$

$$+ \sum_{k'\in BZ'} -\varepsilon(k')\, c_{k'+Q}^+ c_{k'+Q} = \sum_{k\in BZ'} \varepsilon(k)\left(c_k^+ c_k - c_{k+Q}^+ c_{k+Q}\right)$$

$$(BZ' = 1' + 2' + 3' + 4')$$

Description in <u>reduced zone</u> scheme:

$$H_0 = \sum_{k\in BZ} \varepsilon(k)\, c_k^+ c_k = \sum_{k\in BZ'} \varepsilon(k)\left(a_k^+ a_k - b_k^+ b_k\right)$$

where two bands appear:

$$\begin{cases} a_k = c_k & k\in BZ' \\ b_k = c_{k+Q} & k\in BZ' \end{cases}$$

Also $\displaystyle\sum_{k\in BZ} c_{k+Q}^+ c_k = \sum_{k\in BZ'} b_k^+ a_k +$

$$+ \sum_{k\in 1} c_{k+Q}^+ c_k + \sum_{k\in 2} + \sum_{k\in 3} + \sum_{k\in 4}$$

$k = k' + Q$

$k' \in 1'$

$$2\vec{Q} = \frac{2\pi}{a}(1,1) = \vec{0}.$$

$$\sum_{k'\in 1'} c_{k'+2Q}^+ c_{k'+Q} = \sum_{k\in 1'} c_k^+ c_{k+Q} = \sum_{k\in 1'} a_k^+ b_k$$

$$\Rightarrow \sum_{k\in BZ} c^{+}_{k+Q} c_k = \sum_{k\in BZ'} \left(b^{+}_k a_k + a^{+}_k b_k\right)$$

So the whole Hamiltonian reads

$$H_{\uparrow} = \sum_{k\in BZ'} \Big\{ \left(-\varepsilon(k)\right)\left(b^{+}_k b_k - a^{+}_k a_k\right)$$

$$- U m_0 \left(b^{+}_k a_k + a^{+}_k b_k\right) \Big\}$$

$$H_{\downarrow} = H_{\uparrow}\left(m_0 \to -m_0\right)$$

$$H_{\uparrow} = \sum_{k\in BZ'} \left(a^{+}_k, b^{+}_k\right) \begin{pmatrix} \varepsilon_k & -U m_0 \\ -U m_0 & -\varepsilon_k \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix}$$

Eigenvalues $\det \begin{pmatrix} \varepsilon_k - E & -U m_0 \\ -U m_0 & -\varepsilon_k - E \end{pmatrix} = 0$

$$(\varepsilon - E)(-\varepsilon - E) - (U m_0)^2 = 0$$

$$E^2 - \varepsilon^2 - U^2 m_0^2 = 0$$

$$E_k^{(\pm)} = \pm\sqrt{\varepsilon_k^2 + U^2 m_0^2} = \pm E_k .$$

two bands separated by the gap $2 U m_0$.

The ground state energy is

$$E_0 = N U m_0^2 - \sum_{k,\sigma=\uparrow,\downarrow}' E_k = N U m_0^2 - 2 \sum_{k \in BZ'}' E_k$$

because at half-filling the lowest band E_k^- is fully filled and the upper E_k^+ is empty.

Now find m_0 by minimizing E_0:

$$\frac{\partial E_0}{\partial m_0} = 0 \Rightarrow 2 N U m_0 = 2 \sum_k' \frac{\partial E_k}{\partial m_0}$$

$$2 N U m_0 = 2 \sum_k' \frac{2 U^2 m_0}{2\sqrt{\varepsilon_k^2 + U^2 m_0^2}}$$

$$1 = \frac{U}{N} \sum_k' \frac{1}{\sqrt{\varepsilon_k^2 + U^2 m_0^2}}$$

$$\frac{1}{U} = \int_{BZ'} d\vec{k} \frac{1}{\sqrt{\varepsilon_k^2 + U^2 m_0^2}} =$$

$$= \underbrace{\frac{1}{2}}_{BZ' = \frac{1}{2} BZ} \int_{-\lambda}^{\lambda} d\varepsilon \, N(\varepsilon) \frac{1}{\sqrt{\varepsilon^2 + U^2 m_0^2}}$$

$$\frac{1}{U} = \int_0^{\lambda} d\varepsilon \underbrace{\frac{m}{\pi^2} \ln\left(\frac{c}{\varepsilon}\right)}_{N(\varepsilon)} \frac{1}{\sqrt{\varepsilon^2 + U^2 m_0^2}}$$

$\varepsilon = U m_0 x$

$$\frac{1}{U} = \frac{m}{\pi^2}\int_0^{\chi/U m_0} dx \; \frac{U m_0}{U m_0 \sqrt{x^2+1}} \ln\left(\frac{c}{U m_0 x}\right) \sim \frac{m}{\pi^2}\ln^2\left(\frac{c'}{U m_0}\right)$$

hence $$\frac{\pi}{\sqrt{m U}} = \ln\left(\frac{c'}{U m_0}\right)$$

$$\exp\left(\frac{\pi}{\sqrt{m U}}\right) = \frac{c'}{U m_0} \Rightarrow m_0 = \frac{c'}{U}\exp\left(-\frac{\pi}{\sqrt{m U}}\right)$$

1) Order parameter $m_0 \sim c_1 e^{-\frac{c_2}{\sqrt{U}}}$

depends on U in the same way as crit. temperature.

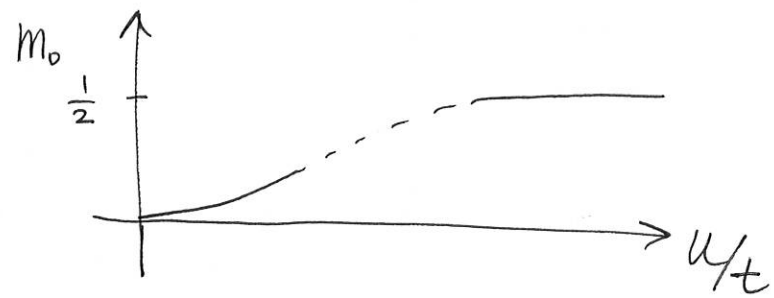
2) Observe that $E_k = \sqrt{\varepsilon_k^2 + U^2 m_0^2}$ is gapped:

$$\min(E_k) = U m_0$$

3) Consider limit of large U now (assuming that mean-field still works):

$$1 = \frac{U}{N}\sum_k{}' \frac{1}{\sqrt{\varepsilon_k^2 + U^2 m_0^2}} \to \frac{U}{N}\sum_k{}' \frac{1}{U m_0} = \frac{1}{2 m_0}$$

$\Rightarrow m_0 = \frac{1}{2}$ makes sence as it describes localized spins $\frac{1}{2}$ arranged antiferromagnetically.



Because of the gap in the spectrum, Um_0, the obtained antiferromagnetic state is <u>insulating</u>.

The insulator is driven by e-e interactions.

This is example of Mott transition: system, which metallic with interactions, turns into insulator by (in this case, arbitrary weak) interaction U.